

Module - III: Fourier Series

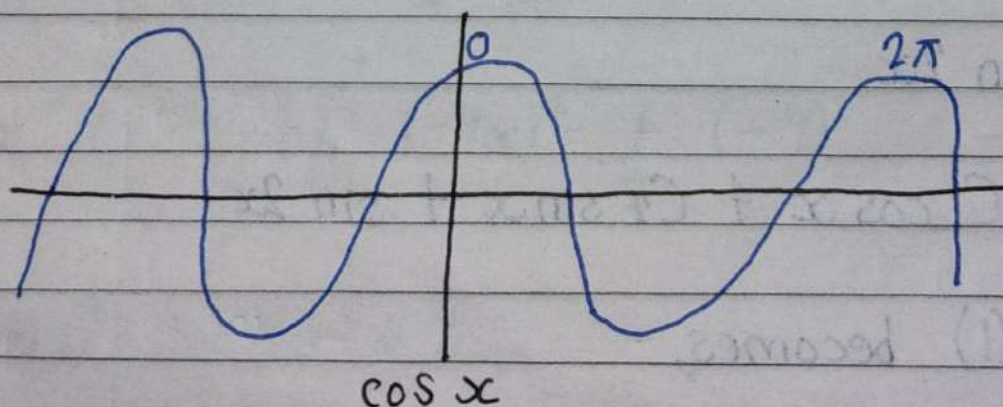
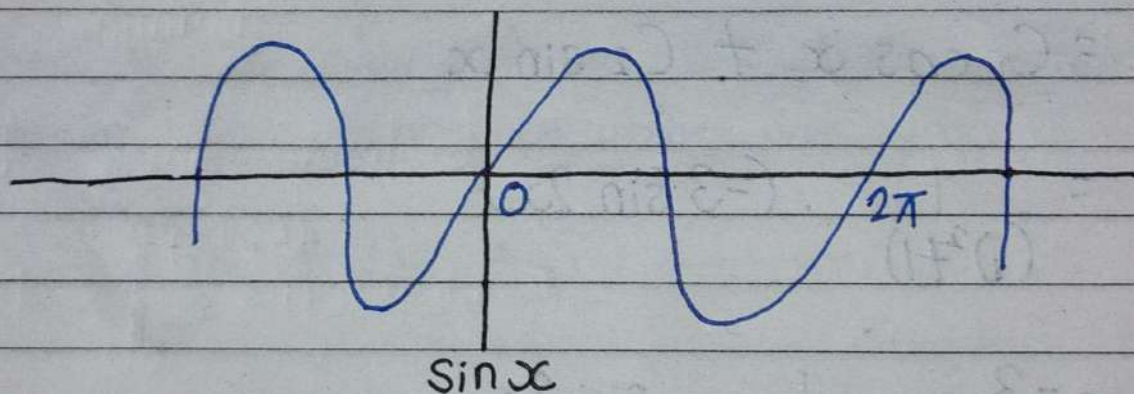
Periodic Function

If the value of each ordinate $f(x)$ repeats itself at equal intervals in the abscissa then $f(x)$ is said to be periodic function.

For example: $\sin x$, $\cos x$, ...

$$\sin x = \sin (x + 2K\pi)$$

$$\cos x = \cos (x + 2K\pi)$$



Fourier Series

A trigonometrical series containing sine & cosine is known as Fourier series.

It is given as,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

where, $a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

Here, $a_0, a_1, a_2, \dots, a_n$ & $b_1, b_2, \dots, b_n$ are constants.

A periodic function can be extended in Fourier series

Dirichlet's Conditions for a Fourier Series

Any function can be extended in fourier series if it satisfy the following conditions i.e. Dirichlet's condition:

1. $f(x)$ is periodic single valued and finite.
2. $f(x)$ has a finite number of discontinuities in any one period.
3. $f(x)$ has a finite number of maxima and minima.

Some Standard Integrals

1. $\int_0^{(\alpha+2\pi)} \sin nx \, dx = 0$
2. $\int_0^{(\alpha+2\pi)} \cos nx \, dx = (-1)^n$
3. $\int_0^{(\alpha+2\pi)} \sin^2 nx \, dx = \pi$
4. $\int_0^{(\alpha+2\pi)} \cos^2 nx \, dx = \pi$
5. $\int_0^{(\alpha+2\pi)} \cos mx \cdot \sin nx \, dx = 0$

$$6. \int_0^{(\alpha+2\pi)} \cos mx \cdot \cos nx \, dx = 0$$

$$7. \int_0^{(\alpha+2\pi)} \sin mx \cdot \sin nx \, dx = 0$$

$$8. \int_0^{(\alpha+2\pi)} \sin mx \cdot \cos nx \, dx = 0$$

By Euler's Formula, the Fourier Series

$$f(x) = \frac{a_0}{2} + a_1 \cos x + a_2 \cos 2x + a_3 \cos 3x + \dots \\ + b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x + \dots$$

Questions

1. Obtain the fourier series of $f(x) = e^{-x}$ in the interval $0 < x < 2\pi$.

Solⁿ:

The fourier series of the function is given as,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$\text{Now, } a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) \, dx = \frac{1}{\pi} \int_0^{2\pi} e^{-x} \, dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-x}}{-1} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} [-e^{-2\pi} + e^0]$$

$$= \frac{1}{\pi} (1 - e^{-2\pi})$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \cdot \cos nx \, dx$$

$$= \frac{1}{\pi} I$$

$$\text{where, } I = \int_0^{2\pi} \underbrace{e^{-x}}_{\textcircled{2}} \underbrace{\cos nx}_{\textcircled{1}} \, dx \quad [\text{Using ILATE}]$$

$$I = \left[\cos nx \int_0^{2\pi} e^{-x} \, dx - \int_0^{2\pi} \left(\frac{d}{dx} \cos nx \int_0^{2\pi} e^{-x} \, dx \right) dx \right]$$

$$= \left[\cos nx \left[\frac{e^{-x}}{-1} \right]_0^{2\pi} - \int_0^{2\pi} -n \sin nx \left[\frac{e^{-x}}{-1} \right] dx \right]$$

$$= \left[-\cos nx \left[e^{-x} \right]_0^{2\pi} - \int_0^{2\pi} n \sin nx \cdot e^{-x} \, dx \right]$$

$$= \left[-\cos(nx) e^{-x} \right]_0^{2\pi} - n \left[\sin nx \frac{e^{-x}}{-1} \right]_0^{2\pi} \\ - \left[-\int_0^{2\pi} n^2 \cos nx \frac{e^{-x}}{-1} dx \right]$$

$$= \left[-\cos(nx) e^{-x} \right]_0^{2\pi} + \left[n \sin(nx) e^{-x} \right]_0^{2\pi} - \left[n^2 \int_0^{2\pi} \cos(nx) e^{-x} dx \right]$$

$$= \left[-\cos(nx) e^{-x} \right]_0^{2\pi} + \left[n \sin(nx) e^{-x} \right]_0^{2\pi} - n^2 I$$

$$\Rightarrow (1+n^2)I = \left[-\cos(2n\pi) e^{-2\pi} + \cos(0) e^0 \right] \\ + \left[n \sin(2n\pi) e^{-2\pi} - n \sin(0) e^0 \right]$$

$$= -1 e^{-2\pi} + 1$$

$$= 1 - e^{-2\pi}$$

$$\therefore I = \frac{1}{1+n^2} [1 - e^{-2\pi}]$$

$$\therefore a_n = \frac{1}{\pi(1+n^2)} [1 - e^{-2\pi}]$$

Now,

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \sin nx \, dx$$

$$= \frac{1}{\pi} I$$

$$I = \int_0^{2\pi} \underset{\textcircled{2}}{e^{-x}} \underset{\textcircled{1}}{\sin nx} \, dx \quad [ILATE]$$

$$= \left[\sin(nx) \frac{e^{-x}}{-1} \right]_0^{2\pi} - \left[\int_0^{2\pi} +n \cos(nx) \frac{e^{-x}}{-1} \, dx \right]$$

$$= \left[-\sin(nx) e^{-x} \right]_0^{2\pi} + \left[n \int_0^{2\pi} \cos(nx) e^{-x} \, dx \right]$$

$$= \left[-\sin(nx) e^{-x} \right]_0^{2\pi} + \left[n \cos(nx) \frac{e^{-x}}{-1} \right]_0^{2\pi}$$

$$- \left[n \int_0^{2\pi} -n \sin(nx) \frac{e^{-x}}{-1} \right]$$

$$= \left[-\sin(nx) e^{-x} \right]_0^{2\pi} - \left[n \cos(nx) e^{-x} \right]_0^{2\pi} - n^2 \int_0^{2\pi} \sin(nx) e^{-x} \, dx$$

$$\Rightarrow I = \left[-\sin(nx) e^{-x} \right]_0^{2\pi} - \left[n \cos(nx) e^{-x} \right]_0^{2\pi} - n^2 I$$

$$\Rightarrow (1+n^2)I = \left[-\sin(2n\pi)e^{-2\pi} + \sin(0)e^0 \right] - n \left[\cos(2n\pi)e^{-2\pi} - \cos(0)e^0 \right]$$

$$\Rightarrow (1+n^2)I = 0 + 0 - ne^{-2\pi} + n$$

$$\Rightarrow (1+n^2)I = n(-e^{-2\pi} + 1)$$

$$\Rightarrow (1+n^2)I = n(1 - e^{-2\pi})$$

$$\Rightarrow I = \frac{n(1 - e^{-2\pi})}{(1+n^2)}$$

$$\therefore b_n = \frac{n(1 - e^{-2\pi})}{\pi(1+n^2)}$$

Hence, the fourier series is,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$= \frac{1}{2\pi} (1 - e^{-2\pi}) + \sum_{n=1}^{\infty} \frac{(1 - e^{-2\pi})}{\pi(1+n^2)} \cdot \cos nx$$

$$+ \sum_{n=1}^{\infty} \frac{n(1 - e^{-2\pi})}{\pi(1+n^2)} \cdot \sin nx$$

$$f(x)$$

$$= \frac{(1-e^{-2\pi})}{2\pi} + \frac{(1-e^{-2\pi})}{\pi} \left[\frac{1}{2} \cos x + \frac{1}{5} \cos 2x + \dots \right] \\ + \frac{(1-e^{-2\pi})}{\pi} \left[\frac{1}{2} \sin x + \frac{2}{5} \sin 2x + \dots \right]$$

2. Find a fourier series to represent $x - x^2$ from $x = -\pi$ to $x = \pi$.

Solⁿ:

The fourier series of the given function be,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

Now,

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x - x^2) dx$$

$$= \frac{1}{\pi} \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{\pi^2}{2} - \frac{\pi^3}{3} - \frac{(-\pi)^2}{2} + \frac{(-\pi)^3}{3} \right]$$

$$= \frac{1}{\pi} \left[\cancel{\frac{\pi^2}{2}} - \frac{\pi^3}{3} - \cancel{\frac{\pi^2}{2}} - \frac{\pi^3}{3} \right]$$

$$= \frac{1}{\pi} \left[-\frac{2\pi^3}{3} \right]$$

$$\Rightarrow a_0 = -\frac{2\pi^2}{3}$$

Then,

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (x - x^2) \cos nx \, dx$$

$$= \frac{1}{\pi} \left\{ \left[\frac{(x - x^2) \sin nx}{n} \right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} dx \cdot \frac{(1 - 2x) \sin nx}{n} \right\}$$

$$= \frac{1}{\pi} \left\{ \left[\frac{(x - x^2) \sin nx}{n} \right]_{-\pi}^{\pi} - \frac{1}{n} \left[\frac{(1 - 2x) \cdot -\cos nx}{n} \right]_{-\pi}^{\pi} + \frac{1}{n} \int_{-\pi}^{\pi} (-2) \cdot \frac{-\cos nx}{n} \, dx \right\}$$

$$= \frac{1}{\pi} \left\{ \left[\frac{(x - x^2) \sin nx}{n} \right]_{-\pi}^{\pi} + \frac{1}{n^2} \left[\frac{(1 - 2x) \cos nx}{n} \right]_{-\pi}^{\pi} + \frac{2}{n^2} \left[\frac{\sin nx}{n} \right]_{-\pi}^{\pi} \right\}$$

$$= \left[\frac{1}{\pi} \frac{(x - x^2) \sin nx}{n} \right]_{-\pi}^{\pi} + \frac{1}{\pi n^2} \left[\frac{(1 - 2x) \cos nx}{n} \right]_{-\pi}^{\pi}$$

$$+ \frac{2}{\pi n^3} \left[\sin nx \right]_{-\pi}^{\pi}$$

$$= \frac{1}{n\pi} \left[\underbrace{(\pi - \pi^2)}_0 \sin n\pi - (-\pi - (-\pi)^2) \sin(-\pi n) \right]$$

$$+ \frac{1}{\pi n^2} \left[(1 - 2\pi) \underbrace{\cos n\pi}_{(-1)^n} - (1 - 2(-\pi)) \cos(-\pi n) \right]$$

$$+ \frac{2}{\pi n^3} [\sin n\pi - \sin(-n\pi)]$$

$$= \frac{1}{n\pi} [0 - 0] + \frac{1}{\pi n^2} [(1 - 2\pi)(-1)^n - (1 + 2\pi)(-1)^n] + 0$$

$$= \frac{1}{\pi n^2} [(1 - 2\pi)(-1)^n - (1 + 2\pi)(-1)^n]$$

$$= \frac{(-1)^n}{\pi n^2} [1 - 2\pi - 1 - 2\pi]$$

$$= \frac{(-1)^n}{\pi n^2} (-4\pi)$$

$$\Rightarrow a_n = \frac{4(-1)^{n+1}}{n^2}$$

$$\text{Also, } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x - x^2) \sin nx \, dx$$

$$= \frac{1}{\pi} \left\{ \int_{-\pi}^{\pi} \left[(x - x^2) \cdot \frac{-\cos nx}{n} \right] - \int_{-\pi}^{\pi} (1 - 2x) \cdot \frac{-\cos nx}{n} \, dx \right\}$$

$$= \frac{1}{\pi} \left\{ \left[\frac{-(x - x^2) \cos nx}{n} \right]_{-\pi}^{\pi} + \frac{1}{n} \left[(1 - 2x) \cdot \frac{\sin nx}{n} \right]_{-\pi}^{\pi} - \frac{1}{n} \int_{-\pi}^{\pi} (-2) \cdot \frac{\sin nx}{n} \, dx \right\}$$

$$= \frac{-1}{n\pi} \left[(x - x^2) \cos nx \right]_{-\pi}^{\pi} + \frac{1}{n^2\pi} \left[(1 - 2x) \sin nx \right]_{-\pi}^{\pi}$$

$$+ \frac{2}{n^2\pi} \left[\frac{-\cos nx}{n} \right]_{-\pi}^{\pi}$$

$$= \frac{-1}{n\pi} \left[(\pi - \pi^2) \cos n\pi - (-\pi - \pi^2) \cos n\pi \right]$$

$$+ \frac{1}{\pi n^2} \left[(1 - 2\pi) \sin n\pi + (1 + 2\pi) \sin n\pi \right]$$

$$- \frac{2}{\pi n^3} \left[\cos n\pi - \cos n\pi \right]$$

$$= \frac{-1}{n\pi} [(\pi - \pi^2)(-1)^n + (\pi + \pi^2)(-1)^n]$$

$$= \frac{(-1)^{n+1}}{n\pi} [\pi - \pi^2 + \pi + \pi^2]$$

$$= \frac{(-1)^{n+1}}{n\pi} (2\pi)$$

$$\Rightarrow b_n = \frac{2(-1)^{n+1}}{n}$$

Hence, the fourier series is,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$= -\frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \cos nx + 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx$$

$$= -\frac{\pi^2}{3} + 4 \left(\cos x - \frac{\cos 2x}{4} + \frac{\cos 3x}{9} - \dots \right)$$

$$+ 2 \left(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \dots \right)$$

3. Find the fourier series of $f(x) = x \sin x$.

Solⁿ:

The fourier series of given function is,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Now,

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} \underset{\textcircled{1}}{x} \underset{\textcircled{2}}{\sin x} dx$$

$$= \frac{1}{\pi} \left[x \cdot -\cos x \right]_0^{2\pi} - \frac{1}{\pi} \int_0^{2\pi} (1) \cdot -\cos x dx$$

$$= -\frac{1}{\pi} \left[x \cdot \cos x \right]_0^{2\pi} + \frac{1}{\pi} \left[\sin x \right]_0^{2\pi}$$

$$= -\frac{1}{\pi} [2\pi \cdot \cos 2\pi - 0] + \frac{1}{\pi} [\sin 2\pi - 0]$$

$$= -\frac{1}{\pi} (2\pi \times 1) + 0$$

$$= -2$$

$$\Rightarrow a_0 = -2$$

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

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Then,

$$a_n = \frac{1}{\pi} \int_0^{2\pi} x \sin x \cdot \cos nx \, dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x \cdot 2 \sin x \cdot \cos nx \, dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x [\sin(x+nx) + \sin(x-nx)] \, dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x \cdot \sin(x+nx) \, dx + \frac{1}{2\pi} \int_0^{2\pi} x \cdot \sin(x-nx) \, dx$$

$$= \frac{1}{2\pi} \left\{ \left[x \cdot \frac{-\cos(n+1)x}{(n+1)} \right]_0^{2\pi} - \int_0^{2\pi} (1) \cdot \frac{-\cos(n+1)x}{(n+1)} \, dx \right\}$$

$$+ \frac{1}{2\pi} \left\{ \left[x \cdot \frac{-\cos(1-n)x}{(1-n)} \right]_0^{2\pi} - \int_0^{2\pi} (1) \cdot \frac{-\cos(1-n)x}{(1-n)} \, dx \right\}$$

$$= \frac{1}{2\pi} \left\{ \frac{2\pi(-1)}{n+1} + \left[\frac{\sin(n+1)x}{(n+1)^2} \right]_0^{2\pi} \right\}$$

$$+ \frac{1}{2\pi} \left\{ \frac{2\pi(-1)}{1-n} + \left[\frac{\sin(1-n)x}{(1-n)^2} \right]_0^{2\pi} \right\}$$

$$= \frac{1}{2\pi} \left\{ \frac{-2\pi}{n+1} + 0 \right\} + \frac{1}{2\pi} \left\{ \frac{-2\pi}{1-n} + 0 \right\}$$

$$= \frac{1}{2\pi} \left[-2\pi \left(\frac{1}{n+1} + \frac{1}{1-n} \right) \right]$$

$$= - \left(\frac{1 - \cancel{n} + \cancel{n} + 1}{1 - n^2} \right)$$

$$= \frac{-2}{1 - n^2}$$

$$\Rightarrow a_n = \frac{2}{n^2 - 1} \quad [n \neq 1]$$

Also,

$$b_n = \frac{1}{\pi} \int_0^{2\pi} x \cdot \sin x \cdot \sin nx \, dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x \cdot 2 \sin x \cdot \sin nx \, dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x \cdot [\cos(x - nx) - \cos(x + nx)] \, dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x \cos(1 - n)x \, dx - \frac{1}{2\pi} \int_0^{2\pi} x \cos(1 + n)x \, dx$$

$$= \frac{1}{2\pi} \left\{ \left[\frac{x \cdot \sin(1 - n)x}{(1 - n)} \right]_0^{2\pi} - \int_0^{2\pi} (1) \cdot \frac{\sin(1 - n)x}{(1 - n)} \, dx \right\}$$

$$- \frac{1}{2\pi} \left\{ \left[\frac{x \cdot \sin(1 + n)x}{(1 + n)} \right]_0^{2\pi} - \int_0^{2\pi} (1) \cdot \frac{\sin(1 + n)x}{(1 + n)} \, dx \right\}$$

$$= \frac{1}{2\pi} \left\{ \frac{2\pi \cdot \sin(1-n)2\pi}{(1-n)} - \left[\frac{-\cos(1-n)x}{(1-n)^2} \right]_0^{2\pi} \right\}$$

$$- \frac{1}{2\pi} \left\{ \frac{2\pi \cdot \sin(1+n)2\pi}{(1+n)} - \left[\frac{-\cos(1+n)x}{(1+n)^2} \right]_0^{2\pi} \right\}$$

$$= \frac{1}{2\pi} \left\{ 2\pi \times 0 + \frac{1}{(1-n^2)} - \frac{1}{(1-n)^2} \right\}$$

$$- \frac{1}{2\pi} \left\{ 2\pi \times 0 + \frac{1}{(1+n)^2} - \frac{1}{(1+n)^2} \right\}$$

$$\left[\frac{1}{2\pi} \left(\frac{1}{(1-n^2)} \right) \right] = \frac{1}{2\pi} \times 0 - \frac{1}{2\pi} \times 0$$

$$\therefore b_n = 0 \quad [n \neq 1]$$

Now,

$$a_1 = \frac{1}{\pi} \int_0^{2\pi} x \sin x \cdot \cos x \, dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x \sin 2x \, dx$$

$$= \frac{1}{2\pi} \left\{ \left[\frac{x \cdot -\cos 2x}{2} \right]_0^{2\pi} - \int_0^{2\pi} (1) \cdot \frac{-\cos 2x}{2} \, dx \right\}$$

$$= \frac{1}{2\pi} \left\{ \left[\frac{-2\pi \cdot \cos 4\pi}{2} - 0 \right] + \left[\frac{\sin 2x}{4} \right]_0^{2\pi} \right\}$$

$$= \frac{1}{2\pi} \left\{ -\frac{2\pi}{2} + \frac{\sin \frac{4\pi}{4}}{4} - 0 \right\}$$

$$= \frac{1}{2\pi} \times -\frac{2\pi}{2}$$

$$= -\frac{1}{2}$$

$$a_2 = \frac{2}{2^2 - 1} = \frac{2}{3}$$

$$a_3 = \frac{2}{3^2 - 1} = \frac{2}{8}$$

Also,

$$b_1 = \frac{1}{\pi} \int_0^{2\pi} x \sin x \cdot \sin x \, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x \sin^2 x \, dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x (1 - \cos 2x) \, dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x \, dx - \frac{1}{2\pi} \int_0^{2\pi} \cos 2x \, dx$$

$$= \frac{1}{2\pi} \left[\frac{x^2}{2} \right]_0^{2\pi} - \frac{1}{2\pi} \left[\frac{\sin 2x}{2} \right]_0^{2\pi} \, dx$$

$$= \frac{1}{2\pi} \times \frac{4\pi^2}{2} - \frac{1}{2\pi} \times \frac{\sin 4\pi}{2}$$

$$= \frac{\pi^2}{\pi}$$

$$\therefore b_1 = \pi$$

Hence, the fourier series is,

$$f(x) = x \sin x = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$= -1 - \frac{1}{2} \cos x + \pi \sin x + 2 \left(\frac{1}{3} \cos 2x + \frac{1}{8} \cos 3x \right)$$